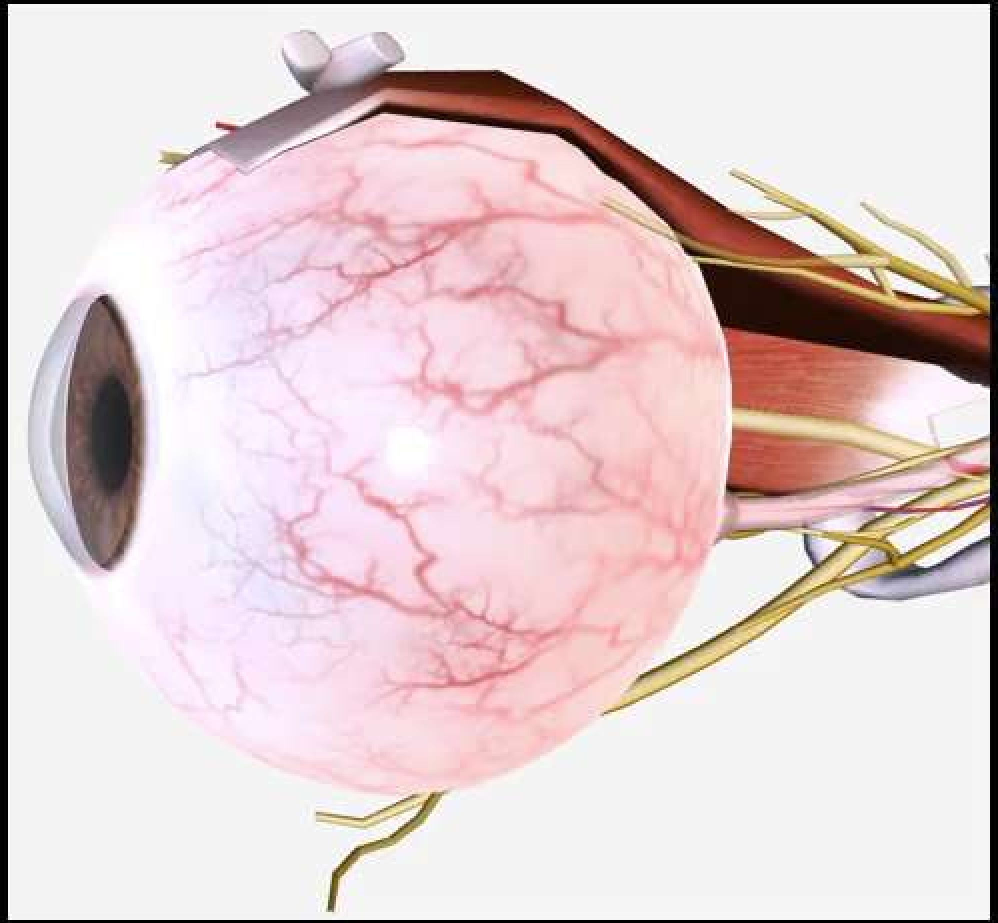
CHAPTER 2 : THE HUMAN EYE AND THE COLOURFUL WORLD

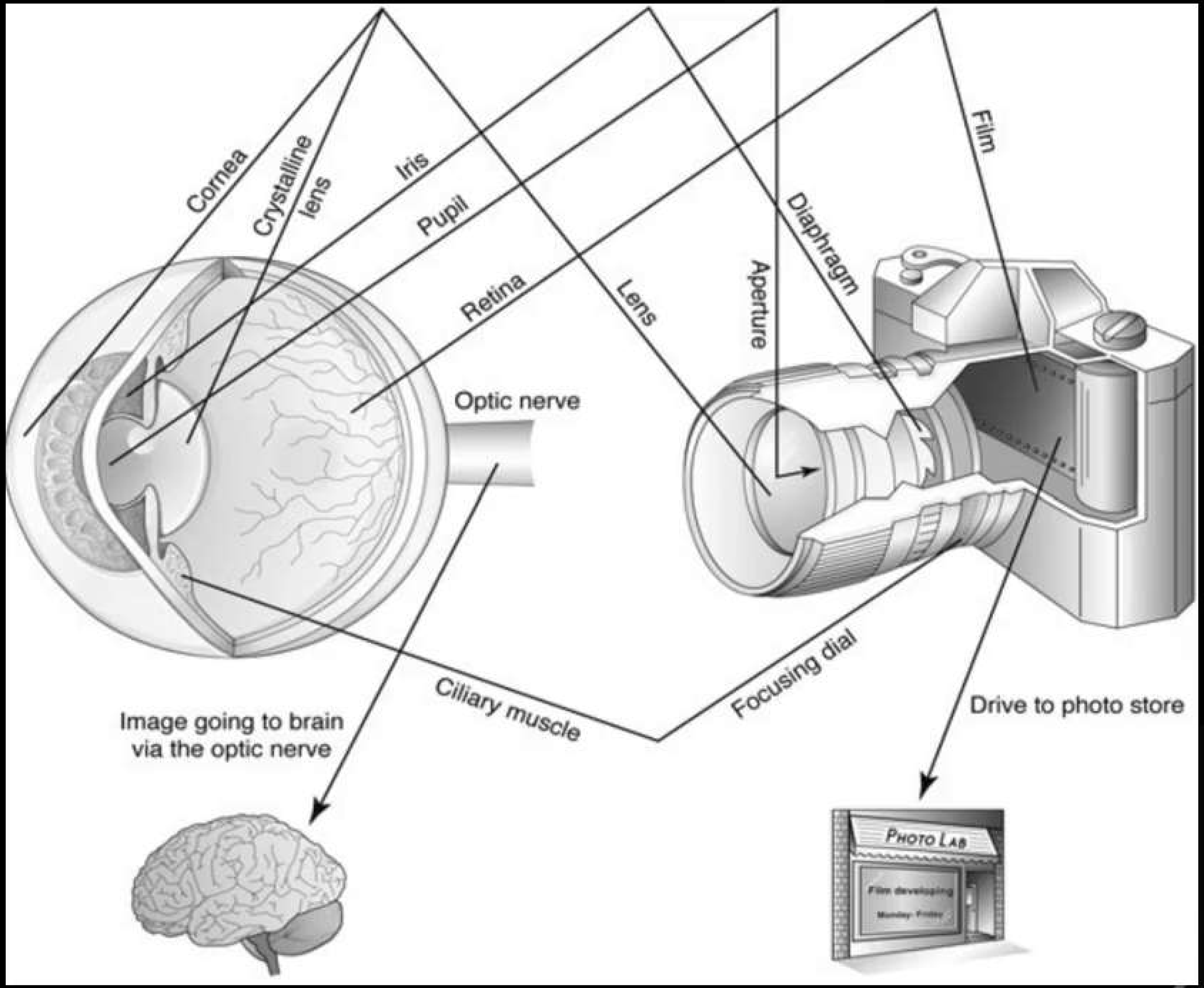
Learning Objectives :

- The Human Eye
- Power of Accommodation
- Defects of Vision and Their Correction
- Refraction of Light Through a Prism
- Dispersion of White Light by a Glass Prism
- Atmospheric Refraction
- Scattering of Light

The Human Eye

- The human **eye** is one of the **most valuable and sensitive sense organs**. It enables us to see the wonderful world and the colours around us.
- The human **eye** is like a **camera**.
- The **eyeball is approximately spherical** in shape with a **diameter** of about **2.3 cm**.





Parts and Function of Human Eye

- **Cornea :**

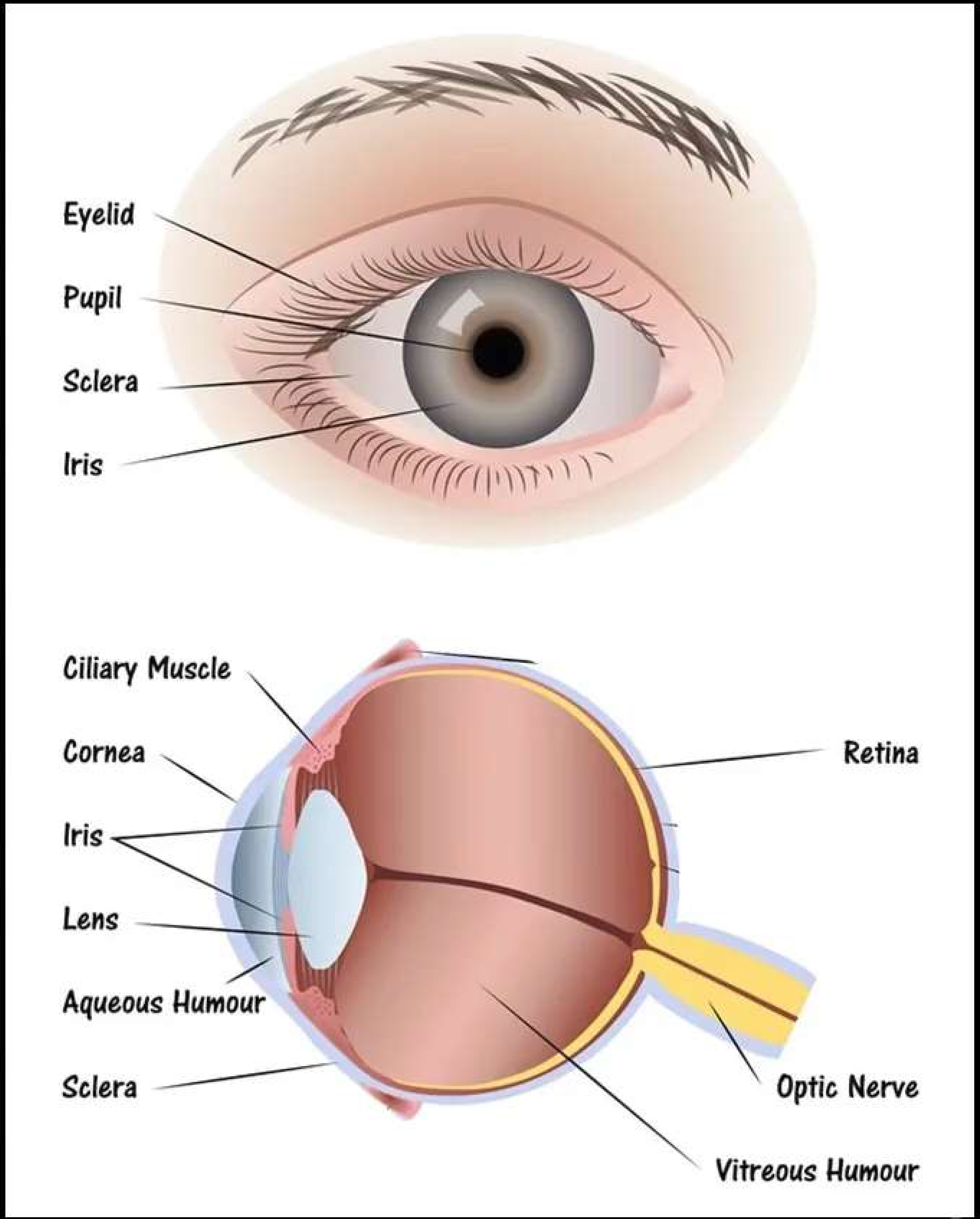
It is a **thin transparent bulged membrane** on the front of the eyeball. **Most of the refraction for the light** entering the eye **occurs at the outer surface of the cornea.**

- **Iris :**

Iris is a **dark muscular diaphragm** and coloured part of the eye behind the cornea, that **controls the size of the pupil.**

- **Pupil :**

The **black, small opening (hole)** in the centre of iris. The **pupil regulates and controls the amount of light entering the eye.**



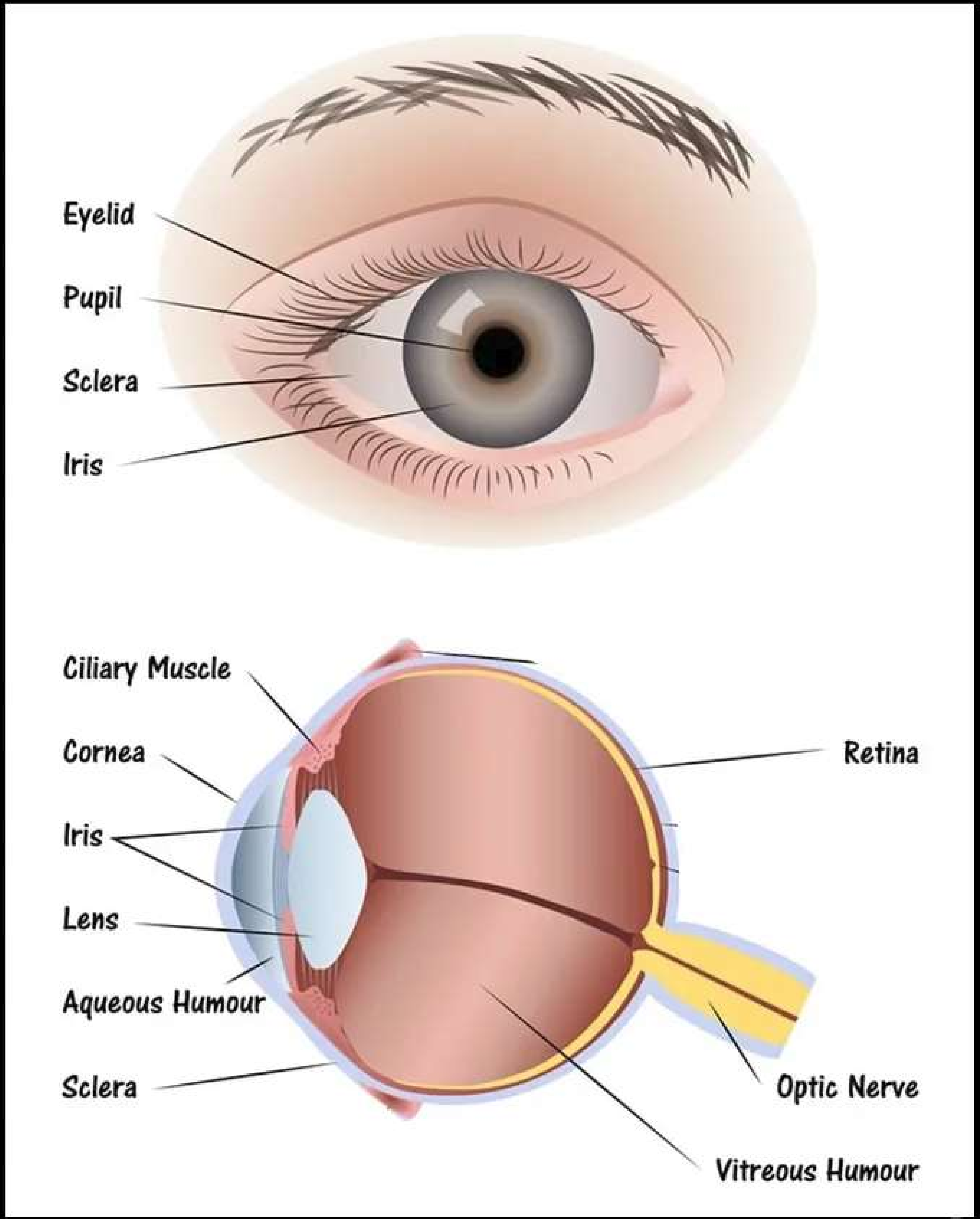
Parts and Function of Human Eye

- **Crystalline lens :**

  It is a **biconvex lens**, composed of **fibrous, jelly like material. It forms an inverted real image of the object on the retina.**
  The **shape/focal length of the lens can be adjusted** with the help of **ciliary muscles** to focus objects at different distances on the retina.

- **Retina :**

  The retina is a **delicate membrane** having enormous number of **light-sensitive cells (Rods and Cones).** The light-sensitive cells get activated upon illumination and **generate electrical signals.** These **signals are sent to the brain via the optic nerves**



Power of Accommodation

- **Accommodation :**

The **ability** of the **eye lens** to **adjust its focal length** is called accommodation..

- **Role of Ciliary muscles :**

The **ciliary muscles are attached to the crystalline lens** and **change its focal length/curvature by contracting and relaxing.**

- When the muscles are **relaxed**, the lens becomes **thin**. Thus, its **focal length increases**. This enables us to see **distant objects** clearly.
- When you are **looking at objects closer** to the eye, the **ciliary muscles contract**.This **increases the curvature** of the eye lens. The eye lens then becomes **thicker**. Consequently, the **focal length of the eye lens decreases**
- However, the **focal length of the eye lens cannot be decreased below a certain minimum limit.**

Power of Accommodation

- **Near Point :**

The **minimum distance**, at which objects can be seen most **distinctly without strain**, is called the **least distance of distinct vision**. It is also called the **near point of the eye**. For a young adult with normal vision, the near point is about **25 cm**.

- **Far Point:**

The **farthest point** upto which the eye can **see objects clearly** is called the **far point of the eye**. It is **infinity** for a normal eye.

- **Cataract:**

Sometimes, the crystalline **lens** of people at **old age becomes milky and cloudy**. This condition is called **cataract**. This **causes partial or complete loss of vision**. It is possible to restore vision through a **cataract surgery**.

Defects of Vision and Their Correction

- **Defects of Vision (Refractive defects):**

Sometimes, the eye is unable to **see distant or nearby objects clearly**. This is due to the **gradually lose its power of accommodation**. These defects can be corrected using **suitable lenses.**

- **Three types of eye defects:**

(1) Myopia (Near-sightedness).

(2) Hypermetropia (Far-sightedness)

(3) Presbyopia

Myopia (Near-sightedness)

- **Myopia :**

Myopia is a **defect of vision** in which a person **can see nearby objects clearly** but **cannot see distant objects clearly**. A myopic person has the **far point nearer than infinity**.

- **Causes of Myopia :**

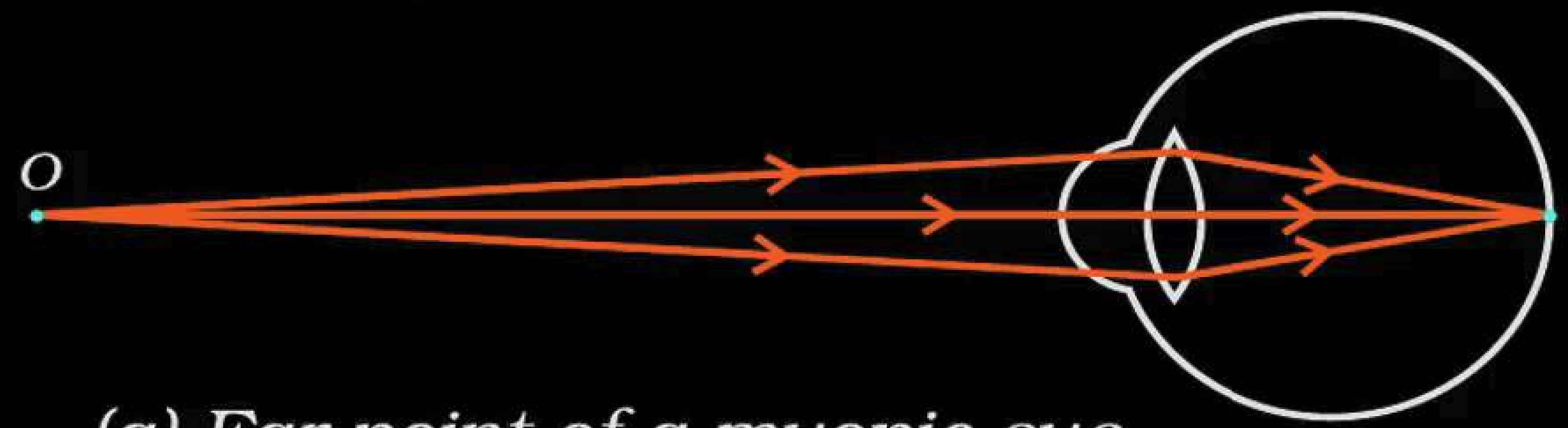
(i) The eye lens becomes **too curved** (focal length decreases)

(ii) The eyeball is **elongated (too long)**.
This causes the **image** to be formed in **front of the retina**.

- **Correction of Myopia :**

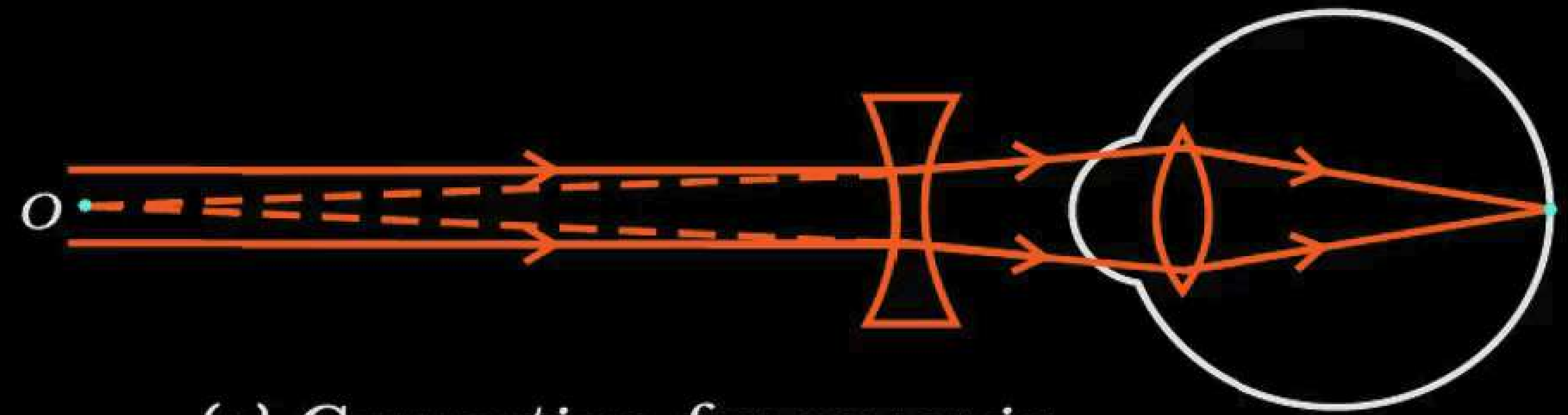
Myopia can be corrected by using a **concave lens** of suitable focal length. The concave lens forms an image back on to the retina.

Myopia (Near-sightedness)



(a) Far point of a myopic eye

(b) Myopic Eye



(c) Correction for myopia

Hypermetropia (Far-sightedness)

- **Hypermetropia :**

Hypermetropia is a **defect of vision** in which a person **can see distant objects clearly** but **cannot see nearby objects clearly**. A hypermetropic person has the **near point farther than 25 cm**.

- **Causes of Hypermetropia :**

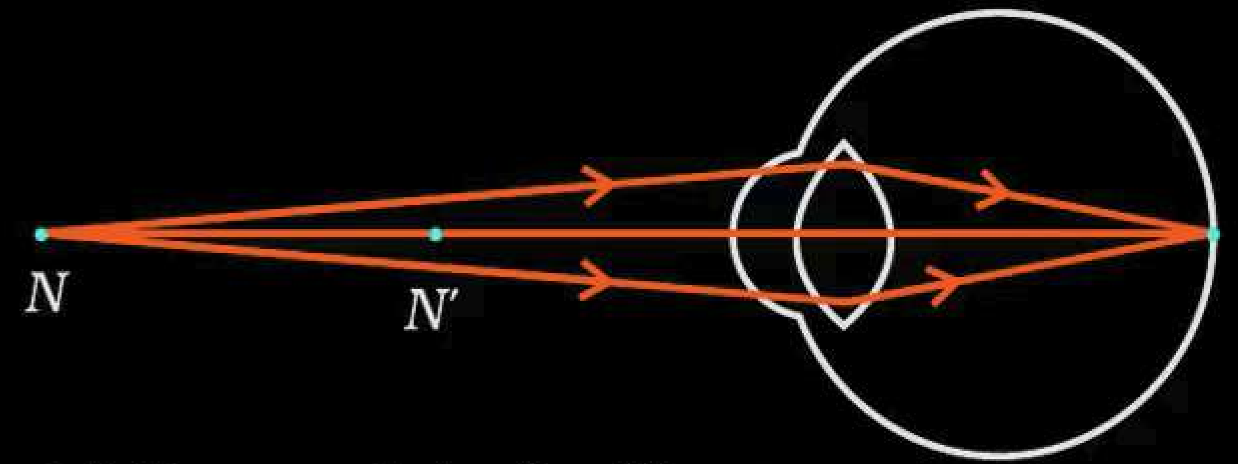
(i) The eye lens **focal length increases)**

(ii) The eyeball is shortened (too small).
This causes the image to be formed behind the retina.

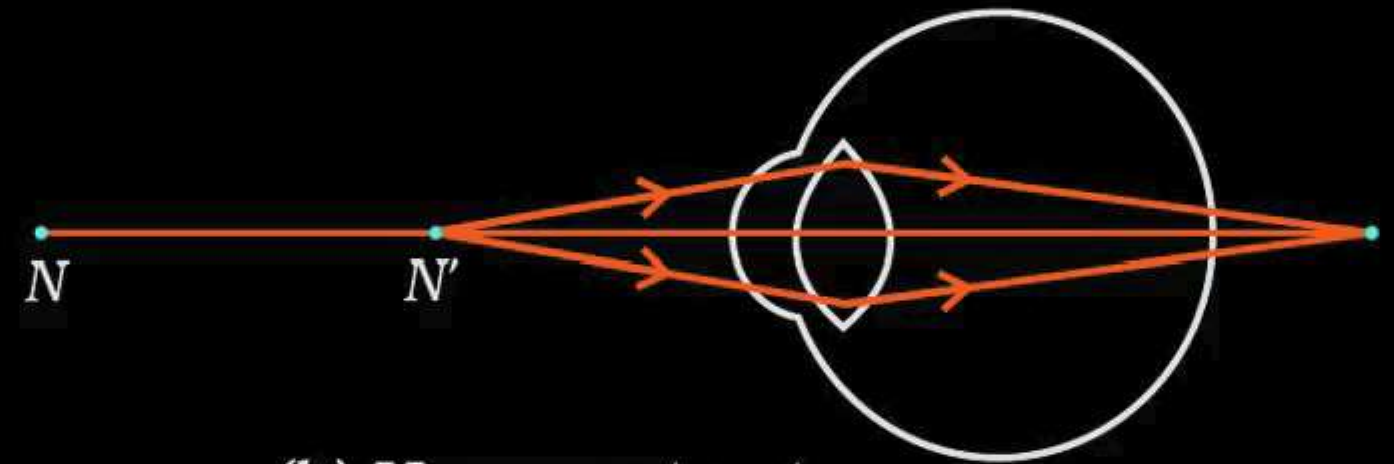
- **Correction of Hypermetropia :**

Hypermetropia can be corrected by using a **convex lens** of suitable focal length. The convex lens forms an image back on to the retina.

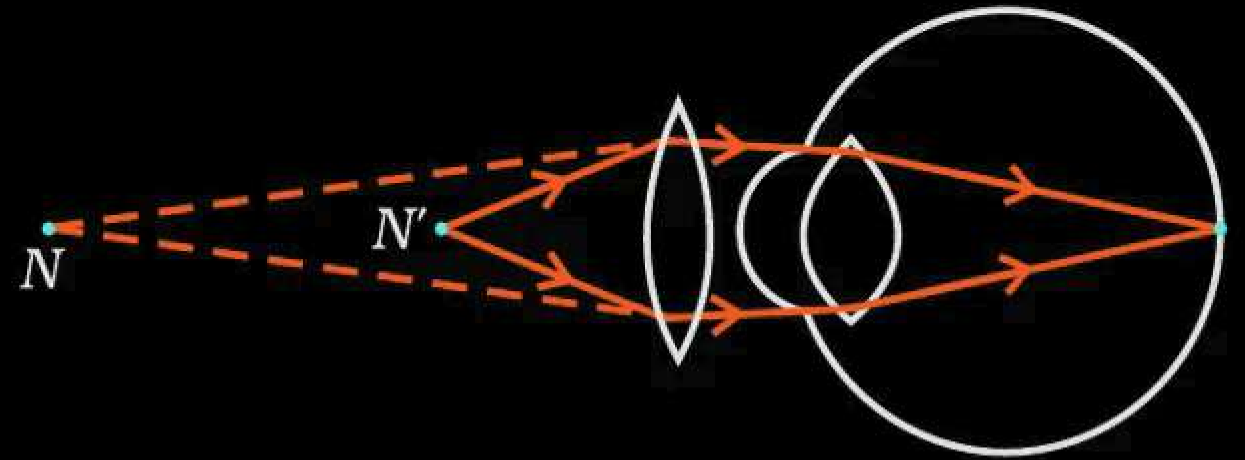
Hypermetropia (Far-sightedness)



(a) Near point of a Hypermetropic eye



(b) Hypermetropic eye



(c) Correction for Hypermetropic eye

Presbyopia

- **Presbyopia :**

Presbyopia is a **defect of vision** in which a person **cannot see nearby objects clearly** and **some person can also cannot see distant objects clearly**. It is caused due to the **gradual weakening of ciliary muscles** and **diminishing flexibility of the eye lens** with age.

- **Correction of Presbyopia :**

Presbyopia can be corrected by using a **bifocal lens** consists of both concave and convex lenses. The **upper portion** consists of a **concave lens**. It facilitates **distant vision**. The **lower part** is a **convex lens**. It facilitates **near vision**.

Presbyopia


Bifocal Lens
Concave lens
for Distant vision
Convex lens
for Near vision

Refraction of Light Through a Prism

- **Triangular Glass Prism :**

  It has **two triangular bases** and **three rectangular** lateral surfaces. These surfaces are inclined to each other.

- **Angle of the prism :**

  The **angle between its two lateral faces** is called the angle of the prism.

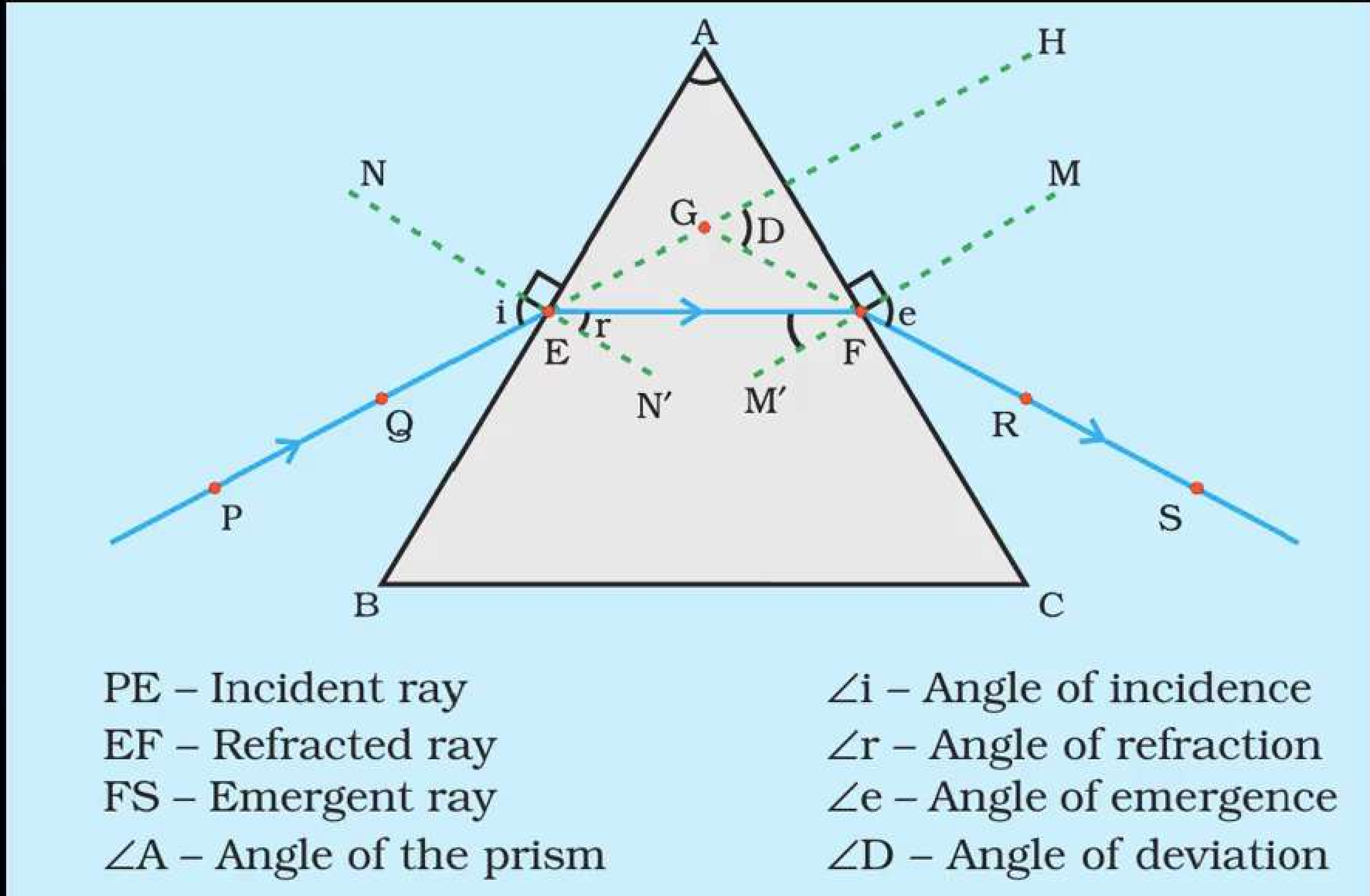


Figure 11.4 *Refraction of light through a triangular glass prism*

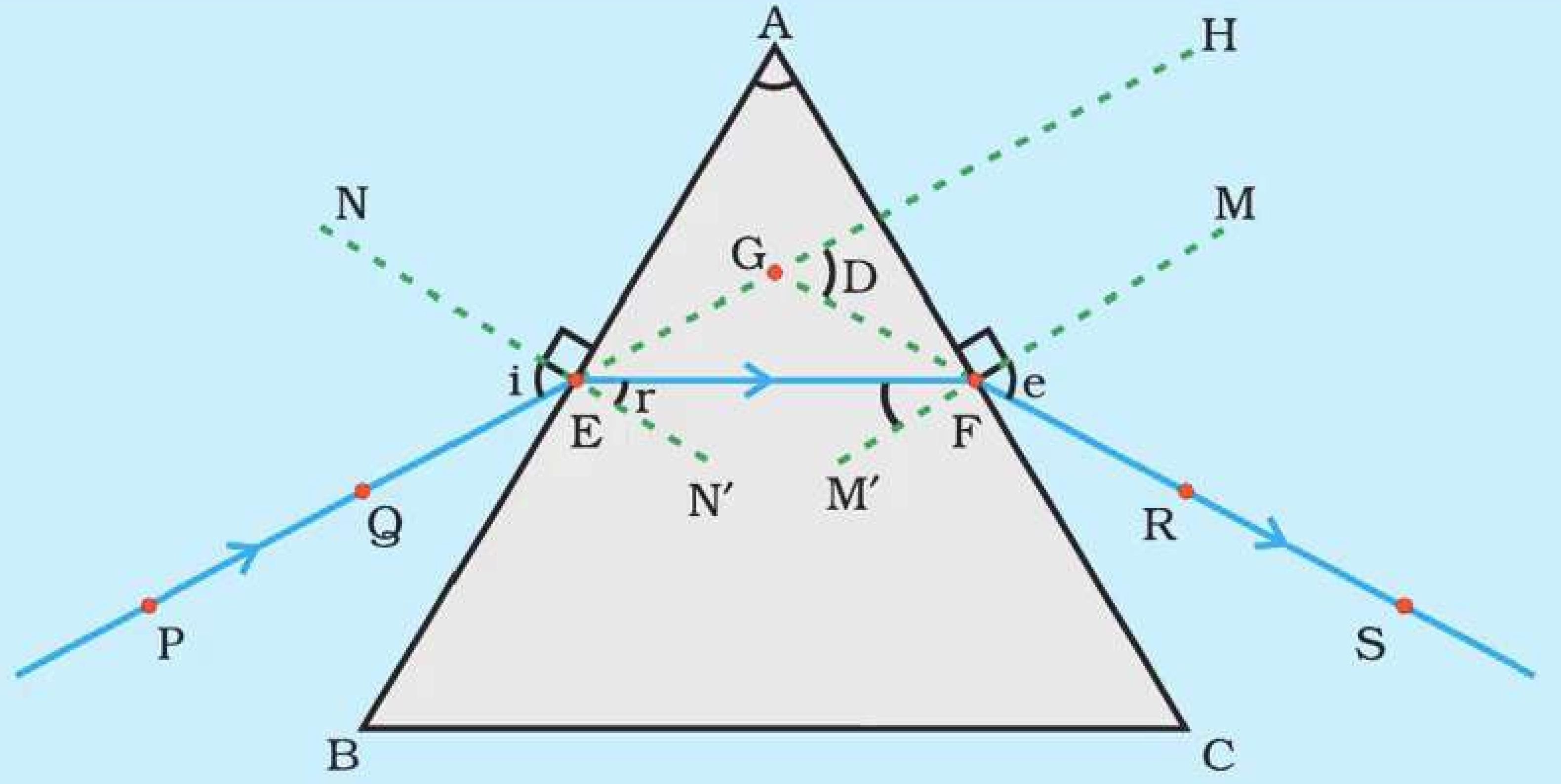
- **Refraction of light by prism :**

  A ray of light is entering **from air to glass** at the **first surface AB**. The light ray on refraction has **bent towards the normal**. At the **second surface AC**, the light ray has **entered from glass to air**. Hence it has **bent away from normal**.

Refraction of Light Through a Prism

- **Angle of Deviation (∠D) :**

The **peculiar shape of the prism** makes the emergent ray bend at an angle to the direction of the incident ray. This **angle between the incident ray and emergent ray** is called the angle of deviation.



PE – Incident ray
EF – Refracted ray
FS – Emergent ray
∠A – Angle of the prism

∠i – Angle of incidence
∠r – Angle of refraction
∠e – Angle of emergence
∠D – Angle of deviation

***Figure 11.4** Refraction of light through a triangular glass prism*

Dispersion of White Light by a Glass Prism

- **Dispersion of light :**

  The **splitting of light into its constituent colours** is called dispersion.

- **Formation of Spectrum :**

  When a beam of white light is incident on a prism, it gets **dispersed into its constituent colours (VIBGYOR)**

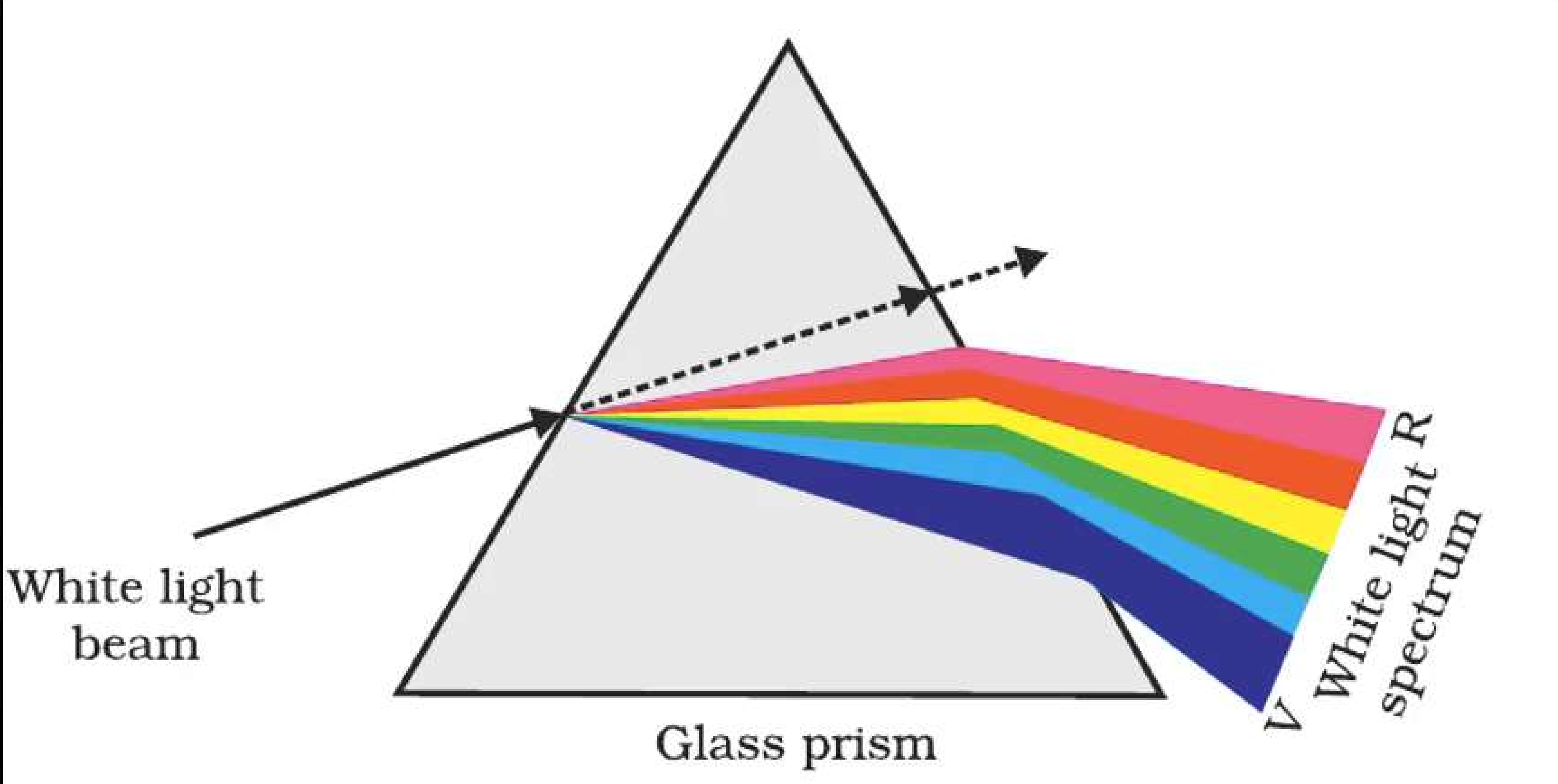


Figure 11.5 *Dispersion of white light by the glass prism*

- **Reason :**

  Different colours of light bend through different angles as they pass through a prism. The **violet colour bends the most** and **red colour bends the least**.

- **Spectrum :**

  The **band of colours** obtained on the screen is called the **spectrum** of light.

Recombination of Colours

- **Recombination of Colours :**

  Isaac Newton used a glass prism to obtain the spectrum of sunlight. He then placed a **second identical prism** in an **inverted position** with respect to the first prism.

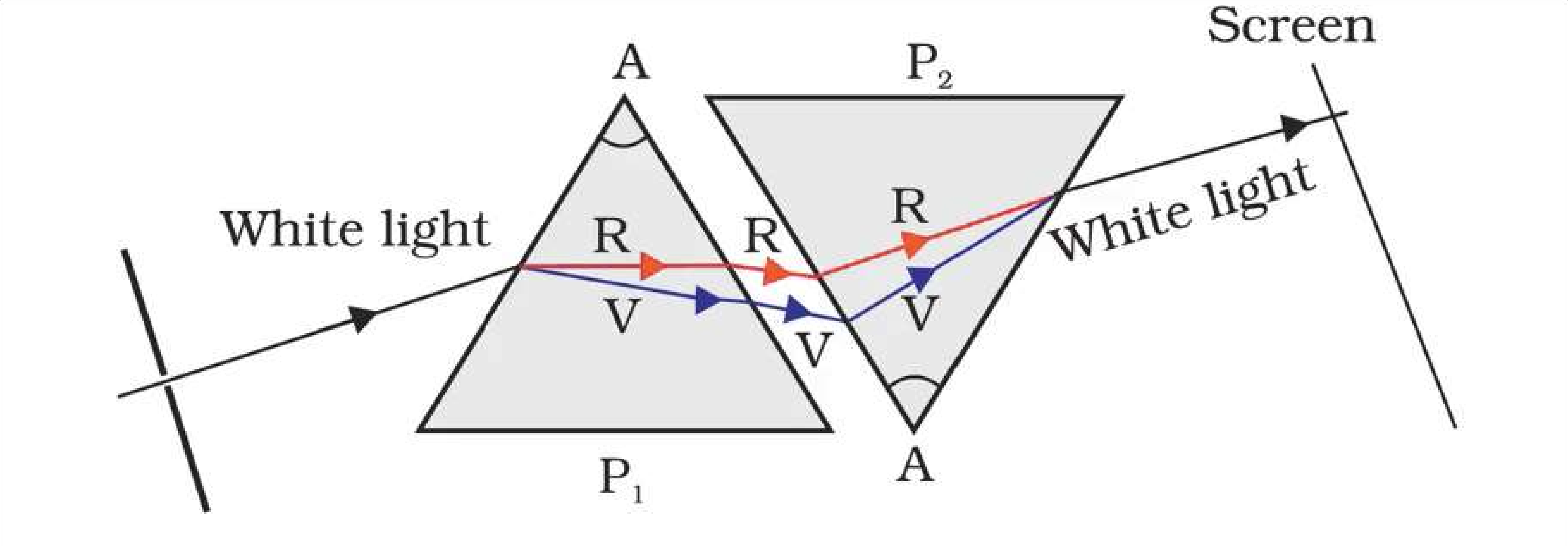


Figure 11.6 *Recombination of the spectrum of white light*

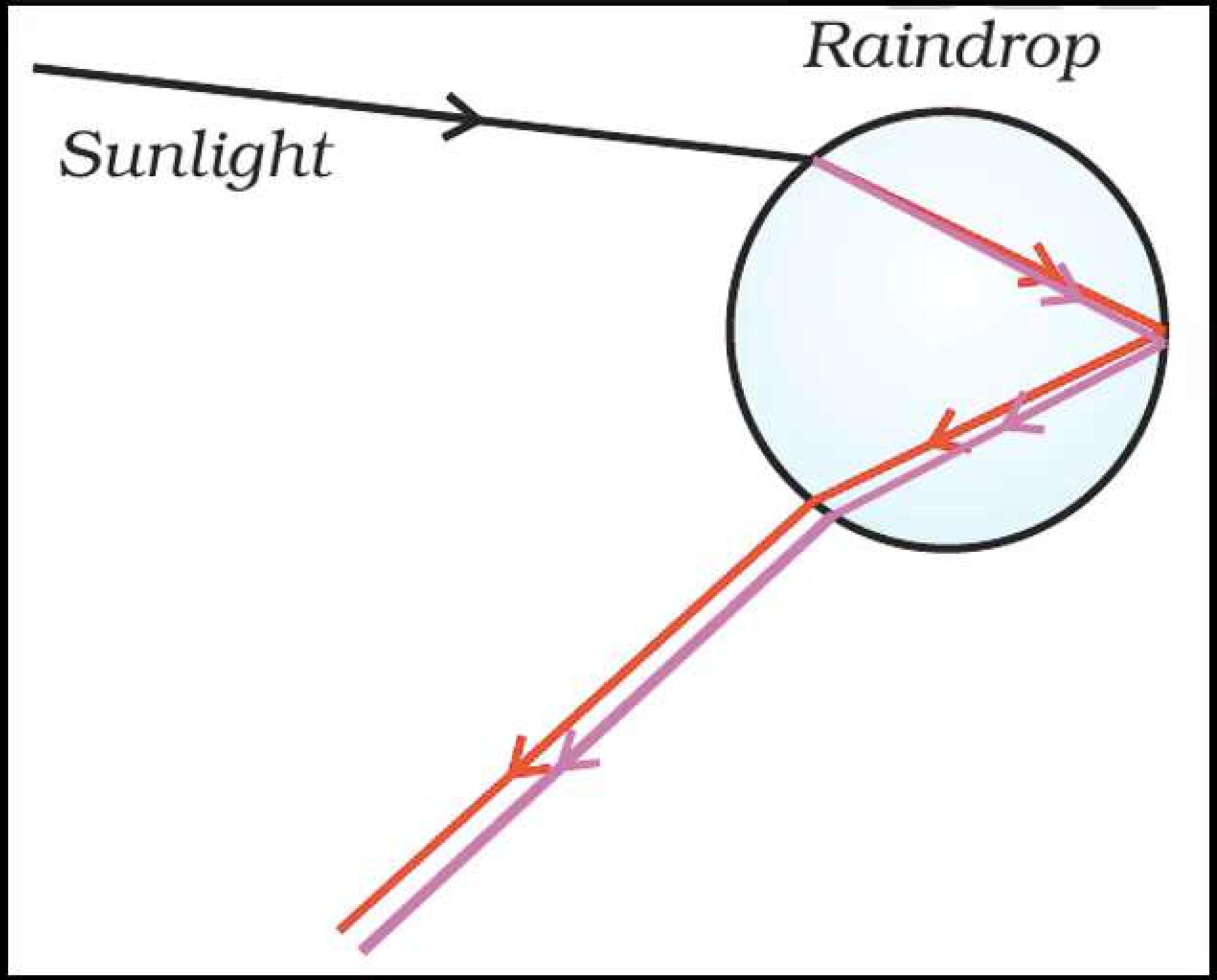
- **Formation of White Light :**

  All the colours of the spectrum **from** the **first prism when passes through the second prism**. He **found a beam of white light emerging** from the other side of the **second prism**.

Rainbow

- **Rainbow :**

A rainbow is a **natural spectrum** appearing in the sky **after a rain shower**. A **rainbow** is always **formed** in a direction **opposite to** that of the **Sun**.



- **Rainbow Formation :**

The **water droplets act like small prisms**. They **refract and disperse** the incident sunlight, **then reflect it internally**, and **finally refract it again** when it comes out of the raindrop. Due to the dispersion of light and internal reflection, different colours reach the observer's eye.

- **Other places where we can see rainbow :**

You can also see a rainbow near a **waterfall** or through a **water fountain**, with the Sun behind you.

Atmospheric Refraction

- You might have observed the apparent **random wavering or flickering of objects** seen through a stream of hot air rising above a **fire**.
- The **hotter air is lighter (less dense) than the cooler air** above it, and has a **refractive index slightly less than that of the cooler air**.
- Since the **physical conditions** of the refracting medium (air) are **not stationary**, the **apparent position** of the object, as seen through the hot air, **fluctuates**.

- This wavering is thus an effect of **atmospheric refraction** (refraction of light by the earth's atmosphere) on a small scale in our local environment.

Twinkling of Stars

- The **twinkling of stars** is due to the **atmospheric refraction** of light.
- Since the **atmosphere bends starlight towards the normal**, the **star appears slightly higher (above) than its actual position** when viewed near the horizon.
- Since the **stars are very distant**, they approximate **point-sized sources of light**.
- Since the **physical conditions** of the earth's **atmosphere are not stationary**. The **refractive index** of air is **not constant** and changes with the change in density.
- This causes the apparent position of the star to change slightly and rapidly and the **amount of starlight entering the eye flickers**- the star sometimes appears **brighter**, and at some other time, **fainter**, which is the twinkling effect.

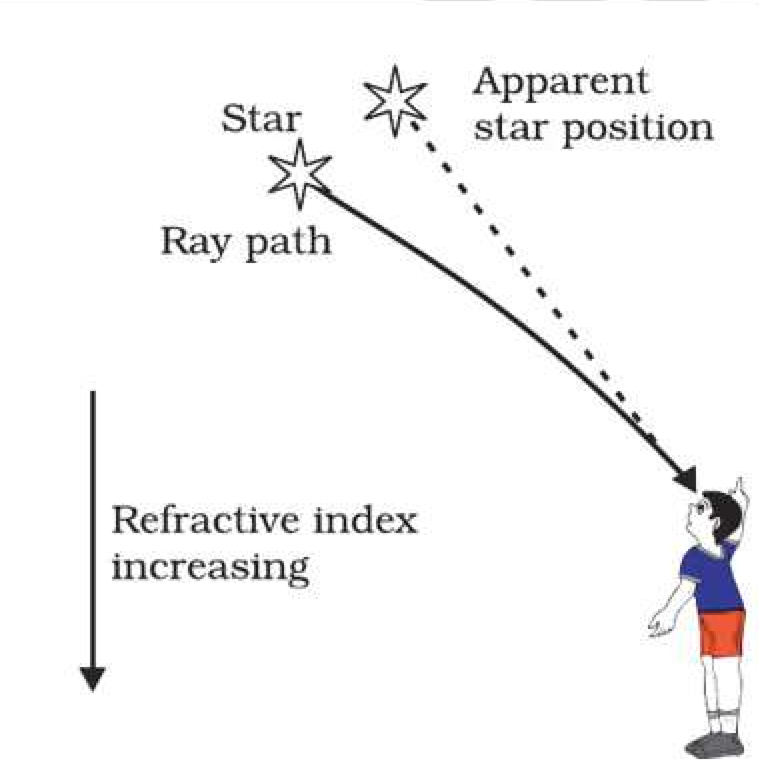


Figure 11.9
Apparent star position due to atmospheric refraction

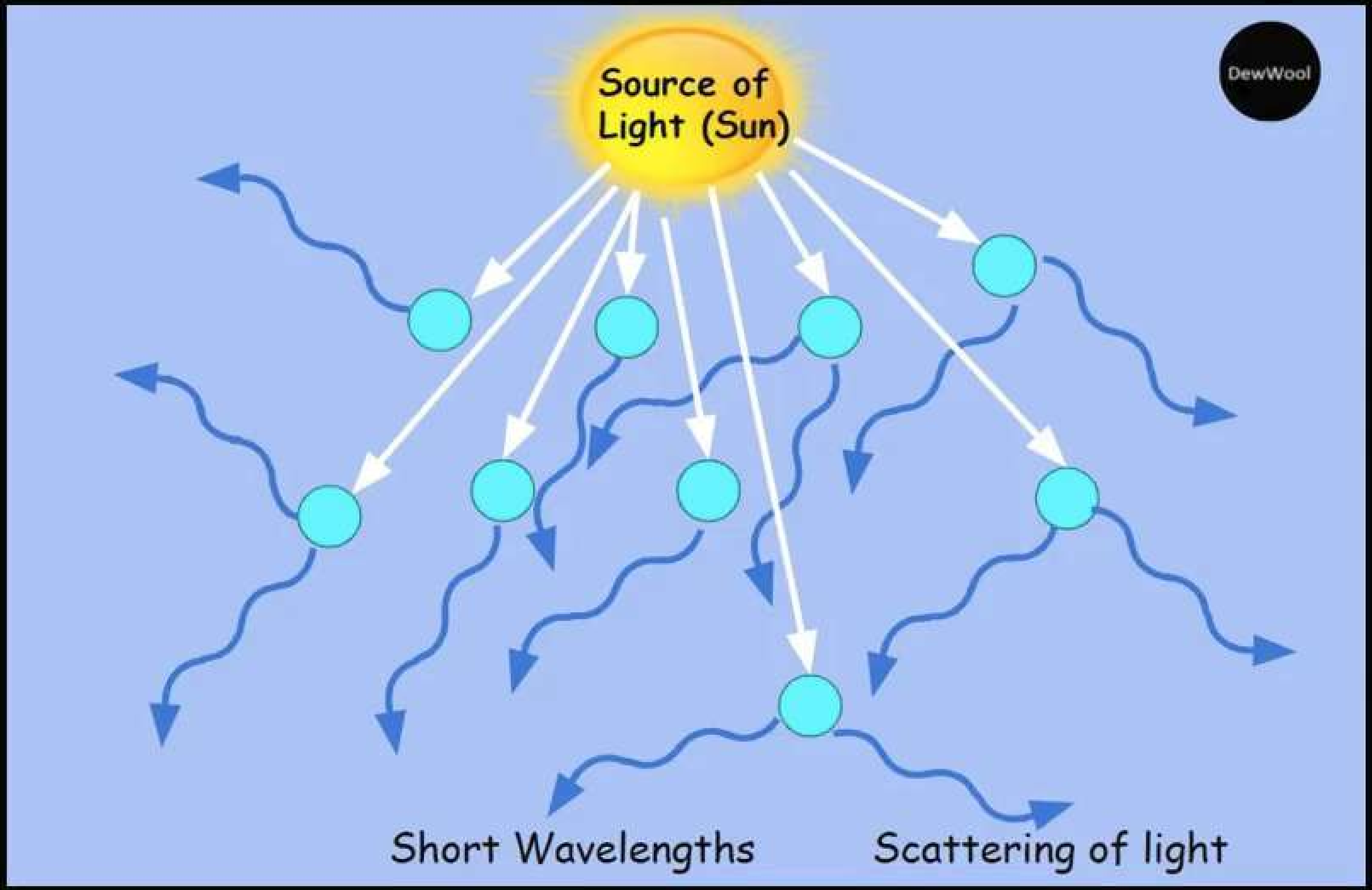
Twinkling of Stars

- **Planets** are **closer to the Earth** than stars, and hence they appear as **extended sources of light**.
- If we consider a **planet** as a **collection of a large number of point-sized sources of light**, the **total variation in the amount of light entering our eye** from all the individual point-sized sources will **average out to zero**, thereby **nullifying the twinkling effect**.

Scattering of Light and Tyndall Effect

What is Scattering of Light?

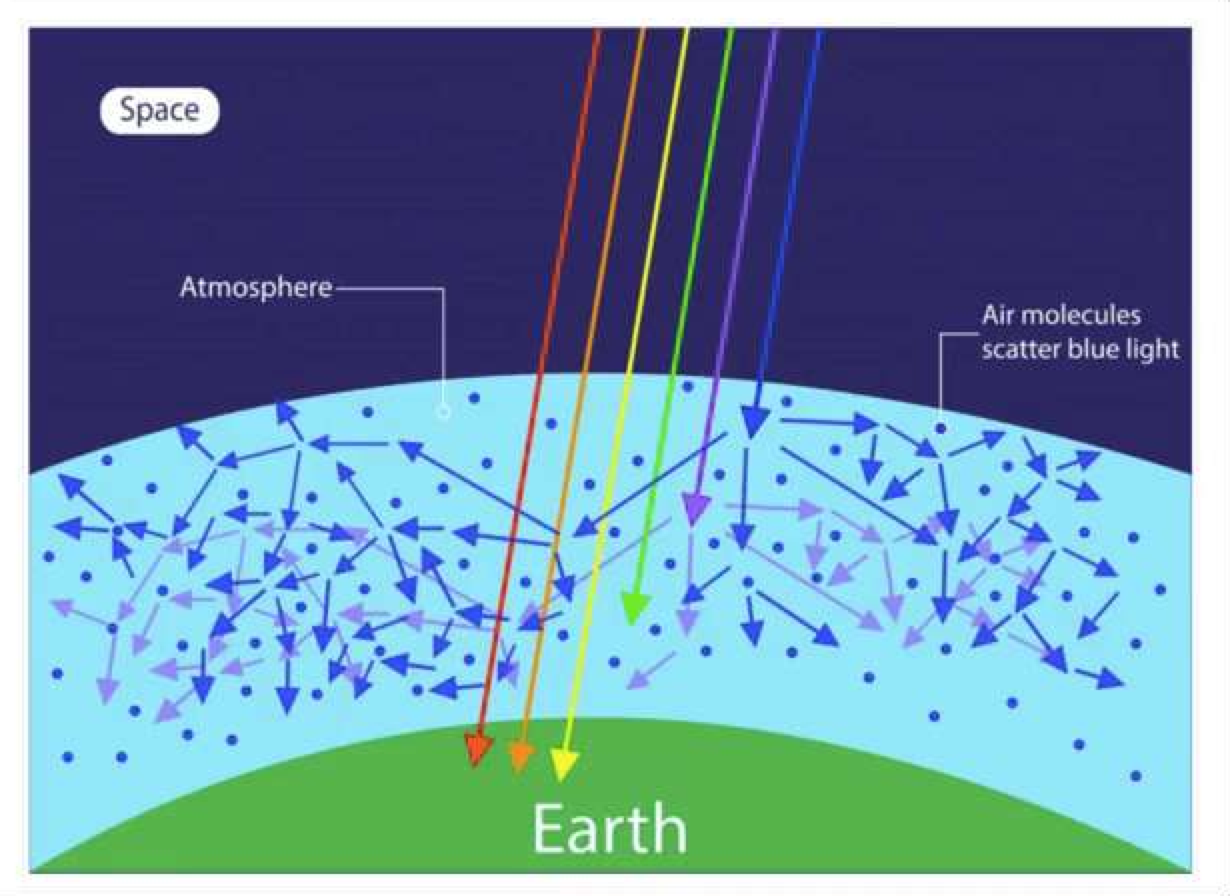
- Light **reflected diffusely** (reflect in all directions) when it strikes very small particles like air molecules or dust.
- This phenomenon is called **scattering**.
- **Shorter wavelengths (e.g., blue)** scatter more than longer ones (e.g., red).
- Explains natural effects like **blue sky** and **reddish sunset**.



Scattering of Light and Tyndall Effect

Why is the Sky Blue?

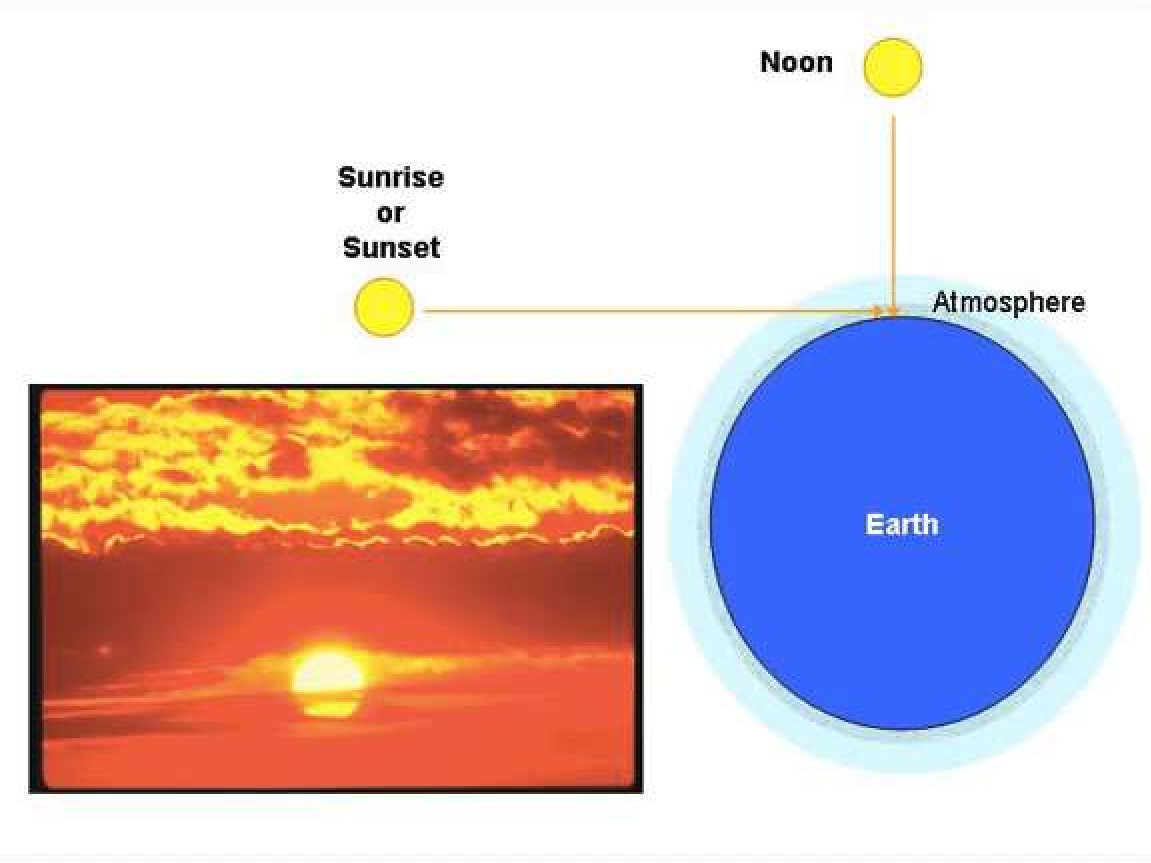
- Sunlight contains all visible wavelengths.
- **Blue light** is scattered more due to its shorter wavelength.
- Our eyes are more sensitive to blue than violet.
- So, the sky appears **blue** to us.



Scattering of Light and Tyndall Effect

Why is the Sun Red at Sunrise and Sunset?

- At sunrise/sunset, sunlight travels a **longer distance** in the atmosphere.
- **Shorter wavelengths (blue)** get scattered out.
- **Red/orange light** reaches our eyes.
- Hence, the Sun appears **reddish**.



Scattering of Light and Tyndall Effect

Tyndall Effect

- **Tyndall effect** is the scattering of light by **colloidal particles**.
- It makes the **path of the beam visible**.
- **Observed in**: fog, dusty rooms, forest mist.
- **Not seen** in pure air or true solutions.